

EXDON EXAM

NAME: _____

Score: _____/15 points

Exercise A: Using the following scenario**10 points**

1. Draw the conceptual model 0.5
2. Identify the possible Entities, their Attributes and the relationships between the entities from the given scenario. 2p
3. Draw the ERD following diagramming convention. 4p
4. Draw the physical model 0.5
5. Write the sentences for each relationship, in both directions. 1p
6. Map the relationships. 0.5
7. Draw the matrix diagram. 1.5

You are a small database consulting company specializing in developing databases for the medical industry. You have just been awarded the contract to develop a data model for a database application system for a mid-size health insurance company to keep track of health claims including patient information, provider(doctor) information, information about patient visits to their doctor as well as prescription drugs prescribed to patients.

Information such as patient name, address, phone, email etc. are needed as well as who each patient's primary care doctor is, their insurance ID number and insurance company name. We also want information on each doctor such as their specialty and what hospitals they are affiliated with as well as their phone, address etc. Regarding the hospitals themselves we will need to know where they are located and how to contact them.

The prescriptions given to each patient by a health-care provider also need to be tracked in this particular database at this time to determine claim eligibility including some basic information on the drug being prescribed to make sure there are no conflicts with a patient's other prescriptions. We need to know each drug's name, purpose/use and possible side effects.

Eventually, the database will be used to track trends and for some extrapolative modeling based on the accumulated data. The database will be accessible in English only right now, although plans include making it available in multiple languages eventually.

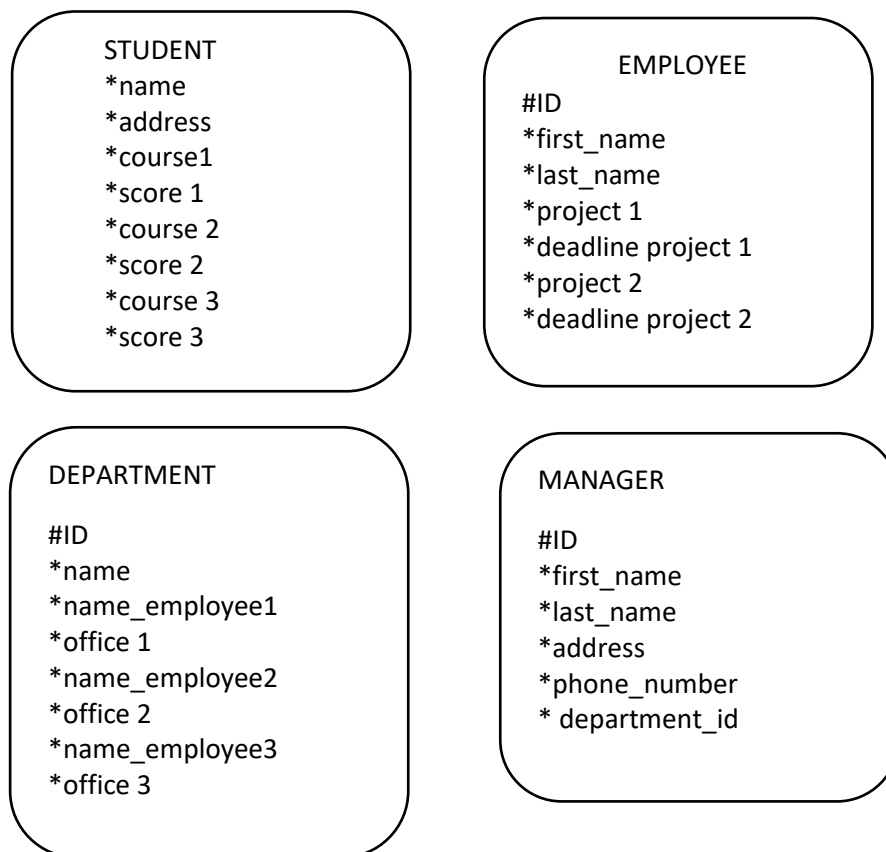
Exercise B: Read the following scenario and construct an ERD that contains at least two subtypes of the entity PRODUCT. Show clearly which attributes belong to the entity supertype, and which belong to the subtypes. Identify a UID for the entity. (1p)

“Our shops sell several kinds of women’s clothing, including dresses, skirts and blouses. Of course each product has a name, a description, and a price. Oh, and sizes too: all products have a waist size. Dresses and skirts have a hem length but blouses don’t. Dresses and blouses have a chest size, but skirts don’t.”

Exercise C: Determine how exclusive relationships should be modeled in the following scenario: (1p)

The Oracle Baseball League store has two types of customer, there are individuals who purchase items like balls, cleats, gloves, shirts, screen printed t-shirts, and shorts. Teams and individual customers are free to purchase any item from the inventory list.

Exercise D: Are the following entities normalized? If not, make the necessary changes to correct it. (2p)



Exercise E: Resolve the M:M relationships.

(1p)

